Three Phase AC-DC Converter

Introduction

The rapid expansion of electric vehicles (EVs) necessitates robust and efficient charging infrastructure. While single-phase AC-DC converters suffice for lower power applications, such as onboard chargers or off-board chargers up to 7 kilowatts, the demand for high-power charging (exceeding 15 kilowatts, often up to 30 kilowatts or more for DC chargers) introduces a critical need for more sophisticated power conversion solutions. These high-power scenarios typically require a three-phase input from the grid, moving beyond the capabilities of single-phase systems.

This lecture delves into the realm of three-phase AC-DC converters, which are essential components in high-power EV charging systems. Unlike their single-phase counterparts, these converters are designed to handle the increased power demands by efficiently converting three-phase alternating current (AC) from the grid into direct current (DC) for battery charging. We will explore the architecture, operational principles, and design considerations for these crucial power electronics, understanding how they enable bidirectional power flow and maintain high power quality.

Core Concepts

Term	Definition	Significance
`Front-End Converter (FEC)`	A three-phase AC-DC converter, often an active rectifier, positioned at the initial stage of a power conversion unit, directly interfacing with the AC grid.	It is crucial for high-power applications, enabling efficient AC-DC conversion and often bidirectional power flow.
`Active Rectifier`	A type of rectifier that uses active switches (like IGBTs or MOSFETs) to control the output voltage and current, allowing for features like unity power factor and bidirectional power flow.	Provides precise control over power conversion, unlike passive rectifiers, which are uncontrolled.

Term	Definition	Significance
`PWM (Pulse Width Modulation)`	A technique where the average value of a voltage or current waveform is controlled by varying the width of the pulses in a train of pulses.	Fundamental to controlling the output voltage and current of the converter, allowing for sinusoidal pole voltage variations.
`Modulation Index (m)`	The ratio of the peak amplitude of the modulating signal to the maximum possible amplitude of the carrier signal in a PWM scheme.	Directly influences the amplitude of the average pole voltage variation, allowing control over the converter's output.
`Unity Power Factor (UPF)`	A condition where the current drawn from the AC source is perfectly in phase with the voltage, resulting in minimal reactive power and maximum real power transfer.	Essential for efficient grid interaction, reducing losses and improving power quality in high-power charging systems.
`Bidirectional Power Flow`	The ability of a converter to transfer power in both directions: from the AC grid to the DC load (charging) and from the DC source back to the AC grid (vehicle-to-grid or regenerative braking).	A key feature for modern EV chargers, enabling advanced functionalities like vehicle-to-grid (V2G) services.
`Phasor Diagram`	A graphical representation of sinusoidal voltages and currents as rotating vectors (phasors) in a complex plane, showing their magnitudes and phase relationships.	Used to visualize and analyze the AC circuit behavior, particularly the relationships between supply voltage, converter voltage, and current, aiding in control strategy development.
`DC Link Voltage (Vo)`	The constant DC voltage maintained across the output capacitor of the AC-DC	A critical parameter that defines the operating voltage for the downstream

Term	Definition	Significance
	converter, which serves as an intermediate bus for subsequent power stages.	DC-DC converter and influences inductor sizing.

Detailed Analysis

1. The Need for High-Power Three-Phase Conversion

For low-power EV charging applications, typically up to 7 kilowatts, single-phase AC-DC converters are commonly employed. These can be found in onboard chargers or smaller off-board units. The power conversion unit in such systems generally consists of a single-phase AC-DC converter followed by an isolated DC-DC converter. However, as power requirements escalate for faster charging, particularly for DC chargers starting from 15 kilowatts and extending beyond 30 kilowatts, a single-phase input becomes insufficient. High-power chargers necessitate a three-phase input from the grid.

When a three-phase input is required, there are two primary approaches for AC-DC conversion:
* **Multiple Single-Phase Converters:** Utilizing three separate single-phase AC-DC converters, each connected to one phase.
* **Integrated Three-Phase Converter:** Employing a single, unified converter designed to directly convert three-phase AC into DC. This lecture focuses on the latter, which is the more commonly used and efficient solution.

2. Architecture of the Three-Phase Front-End Converter (FEC)

The most commonly used three-phase AC-DC converter is often referred to as a "Front-End Converter" (FEC) or an active rectifier. Its circuit topology is characterized by a symmetrical structure, comprising three identical half-bridges.

The key components of this converter include:
* **Three-Phase AC Input:** Connected via three inductors (L), one for each phase (A, B, C). These inductors are typically of the same value. The input voltages are sinusoidal and 120 degrees out of phase, e.g., `Vs,peak sin(wt)`, `Vs,peak sin(wt - 120°)`, `Vs,peak sin(wt - 240°)`.
* **Six Active Switches:** Arranged in three half-bridges (S1-S2, S3-S4, S5-S6). These are active switches, typically IGBTs or MOSFETs, capable of blocking the full DC output voltage (Vo) and allowing current flow in either direction.
* **DC Capacitor (C):** Connected across the output of the half-bridges, providing a

stable DC link voltage (Vo) to the load (RL).
* **Load (RL):** Represents the EV battery or subsequent DC-DC converter stage.

This converter is termed "front-end" because it is the first stage connected to the grid, and it allows for bidirectional power flow. Power can flow from the AC grid to the DC load (charging) or from the DC side back to the AC grid (e.g., for vehicle-to-grid applications or regenerative braking).

3. Operational Principles and PWM Application

The operation of the three-phase AC-DC converter leverages the concept of Pulse Width Modulation (PWM), similar to single-phase converters. By varying the turn-on time (duty ratio) of the switches within each half-bridge (e.g., S1 and S2), the average pole voltage over a carrier cycle can be controlled. The average pole voltage variation is directly proportional to the amplitude of the modulating signal.

In a three-phase system, three modulating signals are used, each sinusoidal and 120 degrees out of phase with respect to each other. By applying these modulating signals, the converter can generate sinusoidal pole voltage variations at its output poles (A, B, C). If these modulating signals vary at the same frequency as the line frequency (e.g., 50 Hz), the average pole voltage variations at points A, B, and C will also be sinusoidal, 120 degrees apart, and at the line frequency. This effectively creates three controllable AC voltage sources at the output of the half-bridges.

4. Single-Phase Equivalent Circuit and Phasor Analysis

To understand the power flow and control, a single-phase equivalent circuit is often used, focusing on the fundamental frequency component. This circuit consists of:
* **Supply Voltage (Vs):** The AC source from the grid.
* **Inductor (L):** The series inductor for each phase. Its impedance at fundamental frequency is `jwL`, where `w = 2πfs` (`fs` is the line frequency).
* **Converter Voltage (V_converter):** The controllable AC voltage generated at the output pole of the half-bridge.

The relationship between these voltages and the current (Is) can be described by Kirchhoff's Voltage Law (KVL). Assuming current flows from the AC source to the converter:
`Vs(t) = V_converter(t) + L * d(Is(t))/dt`

In phasor form, this becomes:
`Vs_phasor = V_converter_phasor + jwL * Is_phasor`
Rearranging for the converter voltage:
`V_converter_phasor = Vs_phasor - jwL * Is_phasor`

A key objective for efficient operation is to achieve **unity power factor (UPF)**. This means the current drawn from the source (Is) should be in phase with the supply voltage (Vs).
* **Phasor Diagram for Forward Power Flow (AC to DC):** If `Vs_phasor` is taken as the reference (0 degrees), then `Is_phasor` will also be at 0 degrees for UPF. The term `jwL * Is_phasor` will be 90 degrees leading `Is_phasor` (and `Vs_phasor`). The phasor diagram shows `Vs_phasor` and `jwL * Is_phasor` as orthogonal vectors, and `V_converter_phasor` is the vector sum of `Vs_phasor` and `-jwL * Is_phasor`. In this configuration, `Vs_phasor` typically leads `V_converter_phasor` by a phase angle `delta`.
* **Phasor Diagram for Reverse Power Flow (DC to AC):** For power to flow from the DC side back to the AC grid, the relationship changes. The KVL becomes `V_converter_phasor = Vs_phasor + jwL * Is_phasor` (assuming `Is_phasor` is now flowing from converter to source). In this case, `V_converter_phasor` leads `Vs_phasor` by the phase angle `delta`.

Control of Power Flow:
* **Direction:** By changing the sign of the phase angle `delta` between `Vs_phasor` and `V_converter_phasor`, the direction of power flow can be reversed (AC to DC or DC to AC).
* **Magnitude:** By adjusting the magnitude of `delta`, the magnitude of the current `Is` (and thus the power) can be controlled. A larger `delta` generally corresponds to a larger voltage drop across the inductor, leading to a larger current.

The magnitude of `V_converter_phasor` is given by `m * Vo / 2`, where `m` is the modulation index and `Vo` is the DC link voltage. This allows the converter to generate the necessary voltage to control current and power flow.

5. Inductor Sizing for the Front-End Converter

The inductor (L) plays a critical role in filtering current harmonics and controlling power flow. Its value needs to be carefully selected. Using the phasor diagram for forward power flow (where `Vs` and `jwL Is` are orthogonal, and `V_converter` is the hypotenuse), we can apply the Pythagorean principle:

`|V_converter_phasor|^2 = |Vs_phasor|^2 + |jwL * Is_phasor|^2`

Substituting the magnitudes:
`(m * Vo / 2)^2 = Vs_peak^2 + (wL * Is_peak)^2`

Rearranging to solve for `L^2`:
`(wL * Is_peak)^2 = (m * Vo / 2)^2 - Vs_peak^2`
`L^2 = ((m * Vo / 2)^2 - Vs_peak^2) / (w^2 * Is_peak^2)`

Therefore, the inductance `L` can be calculated as:
`L = sqrt[ ((m * Vo / 2)^2 - Vs_peak^2) / ((2 * pi * fs)^2 * Is_peak^2) ]`

Where:
* `m`: Modulation index (typically kept between 0.8 and 0.9 for normal operation).
* `Vo`: DC link voltage (a specified value).
* `Vs_peak`: Peak of the phase voltage (from grid specification).
* `fs`: Line frequency (e.g., 50 Hz or 60 Hz).
* `Is_peak`: Peak of the phase current (determined by power level and desired UPF operation).

Design Considerations for Inductor:
* **Voltage Drop:** It is crucial to ensure that the voltage drop across the inductor (at fundamental frequency) remains within 10% of the supply voltage. Exceeding this limit can lead to poor DC bus utilization.
* **RMS Current:** The inductor must be sized to handle the RMS current flowing through it, which for a sinusoidal current is `Is_peak / sqrt(2)`. This ensures the inductor does not saturate or overheat.

The inductor is continuously in conduction throughout the line cycle, making its RMS current rating a significant design parameter. The capacitor sizing will be discussed in subsequent lectures, along with closed-loop control strategies for this converter.

Key Takeaways

- [ ] High-power EV charging (15 kW+) requires three-phase AC-DC converters due to increased power demands.
- [ ] The Front-End Converter (FEC) or active rectifier is a common topology for three-phase AC-DC conversion, featuring three half-bridges and six active switches.
- [ ] PWM is used to control the average pole voltage, enabling the converter to generate controllable sinusoidal AC voltages at its output poles.
- [ ] The converter allows for bidirectional power flow (AC to DC and DC to AC) by changing the phase angle between the supply voltage and the converter's output voltage.
- [ ] Unity Power Factor (UPF) operation is achievable, ensuring the current drawn from the grid is in phase with the voltage for high efficiency.
- [ ] The inductor value is critical for controlling current and power flow, and its sizing depends on the modulation index, DC link voltage, and peak phase voltages/currents.
- [ ] The voltage drop across the inductor at fundamental frequency must be limited to within 10% of the supply voltage for optimal DC bus utilization.

- [ ] The inductor must be rated for the RMS current, which is `Is_peak / sqrt(2)` for sinusoidal current.

Conclusion

The three-phase AC-DC converter is a cornerstone technology for the next generation of high-power EV charging infrastructure. Its ability to efficiently convert three-phase AC to DC, coupled with features like unity power factor operation and bidirectional power flow, makes it indispensable for fast charging and advanced grid integration functionalities such as vehicle-to-grid (V2G). Understanding its architecture, the application of PWM, and the principles of phasor analysis is fundamental for designing robust and high-performance charging systems.

The detailed analysis of inductor sizing highlights the critical interplay between electrical parameters and physical component selection, ensuring both efficient power transfer and system stability. As EV technology continues to evolve, the sophistication and efficiency of these front-end converters will be paramount in supporting the widespread adoption of electric mobility and its seamless integration with the electrical grid. Future discussions will build upon this foundation, exploring capacitor sizing and the implementation of closed-loop control to further optimize converter performance.